Leandro Beraldo e Silva | CV

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Education and Training

University of Arizona (w/ Kate Daniel) — **Postdoc**
Tucson, AZ — *2023-present*
Barbara Pichardo Future Faculty Fellow

University of Michigan (w/ Monica Valluri) — **Postdoc**
Ann Arbor, MI — *2021-2023*

University of Central Lancashire (w/ Victor Debattista) — **Postdoc**
Preston, UK — *2018-2021*

University of São Paulo (w/ Laerte Sodré) — **Postdoc**
São Paulo, Brasil — *2015-2018*
(one-year internship at University of Michigan, w/ Monica Valluri)

University of São Paulo (w/ Laerte Sodré & Marcos Lima) — **Ph.D. Physics**
São Paulo, Brasil — *2011-2015*
(one-year internship at Institut d'Astrophysique de Paris, w/ Gary Mamon)

University of São Paulo (w/ Ivone Albuquerque) — **M.Sc. Physics**
São Paulo, Brasil — *2008-2010*

University of São Paulo — **B.A. Phys. & Astronomy**
São Paulo, Brasil — *2003-2007*

List of publications

Full list: ADS
First few authors: ADS

Refereed publications

28. **Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument** — **AJ**
*DESI collaboration, w/ **LBeS**, AJ 167 2 2024* — *2024*

27. **Characterisation of high velocity stars in the S-PLUS internal fourth data release** — **MNRAS**
*F. Quispe-Huaynasi, F. Roig, V. M. Placco, **LBeS** et al, MNRAS 527 3 2024* — *2024*

26. **Early Growing Supermassive Black Holes Strengthen Bars and Boxy/Peanut Bulges** — **ApJ**
*V. Wheeler, M. Valluri, **LBeS**, S. Dattathri, V. P. Debattista, ApJ 958, 2 2023* — *2023*

25. **Orbital support and evolution of flat profiles of bars (shoulders)** — **ApJ**
***LBeS**, V. P. Debattista, S. R. Anderson, M. Valluri, P. Erwin, K. Daniel, N. Deg, ApJ 955 1 2023* — *2023*

24. **Exploring the Evolution of Massive Clumps in Simulations That Reproduce the Observed Milky Way α-element Abundance Bimodality** — **ApJ**
*B. R. Garver, D. Nidever, V. P. Debattista, **LBeS**, T. Khachaturyants, ApJ 953 2 2023* — *2023*

23. **Chemodynamical Properties and Ages of Metal-Poor Stars in S-PLUS** — **MNRAS**
*Almeida-Fernandes, V. Placco, H. Rocha-Pinto et al, w/ **LBeS**, MNRAS 523 2 2023* — *2023*

22. **The imprint of clump formation at high redshift. II. The chemistry of the bulge** — **ApJ**
*V. P. Debattista, D. Liddicott, O. A. Gonzalez, **LBeS** et al, ApJ 946 2 2023* — *2023*

21. **Overview of the DESI Milky Way Survey** — **ApJ**
*The DESI Milky Way Survey Group (w/ **LBeS**), ApJ 947 1 2023* — *2023*

20. **DESI Observations of the Andromeda Galaxy: Revealing the Immigration History of our Nearest Neighbor** — **ApJ**
*A. Dey, J. R. Najita, S. E. Koposov et al (w/ **LBeS**), ApJ 944 1 2023* — *2023*

19. **The pattern speeds of vertical breathing waves** — **MNRAS *Letters***
*T. Khachaturyants, V. P. Debattista, S. Gosh, **LBeS**, K. Daniel, MNRAS* Letters *517 1 2022* — *2022*

18. **GASTRO library I: the simulated chemodynamical properties of several GSE-like stellar halos** — **ApJ**
*J. A. Amarante, V. P. Debattista, **LBeS**, C. F. P. Laporte, N. Deg, ApJ 937 1 2022* — *2022*

17. **The secular growth of bars revealed by flat (peak+ shoulders) density profiles** — **MNRAS**
*S. R. Anderson, V. P. Debattista, P. Erwin, D. J Liddicott, N. Deg, **LBeS**, MNRAS 513 1642 2022* — *2022*

16. **Bending waves excited by irregular gas inflow along warps** — **MNRAS**
*T. Khachaturyants, **LBeS**, V. P. Debattista, MNRAS 512 3 3500 2022* — *2022*

15. **Tracing birth properties of stars with abundance clustering** — **ApJ**
*B. Ratcliffe, M. Ness, T. Buck, K. Johnston et al (w/ **LBeS**), ApJ 924 2 2022* — *2022*

14. **How stars formed in warps settle into (and contaminate) thick discs** — **MNRAS**
*T. Khachaturyants, **LBeS**, V. P. Debattista, MNRAS 508 2 2350 2021* — *2021*

13. **Recovering the origins of the lenticular galaxy NGC 3115 using multiband imaging** — **MNRAS**
*M. L. Buzzo, A. Cortesi, J. Hernandez-Jimenez et al (w/ **LBeS**), MNRAS 504 2 2146 2021* — *2021*

12. **The relative efficiencies of bars and clumps in driving disc stars to retrograde motion** — **MNRAS**
*K. Fiteni, J. Caruana, J. A. S. Amarante, V. P. Debattista, **LBeS**, MNRAS 503 1 1418 2021* — *2021*

11. **Co-formation of the thin and thick discs revealed by APOGEE-DR16 and Gaia-DR2** — **MNRAS**
***LBeS**, V. P. Debattista, D. Nidever, J. A. S. Amarante, B. Garver, MNRAS 502 260 2021* — *2021*

10. **Box/peanut-shaped bulges in action space** — **MNRAS**
*V. P. Debattista, D. J. Liddicott, T. Khachaturyants, **LBeS**, MNRAS 498 3334 2020* — *2020*

9. **Geometric properties of galactic discs with clumpy episodes** — **MNRAS**
***LBeS**, V. P. Debattista, T. Khachaturyants, & D. Nidever, MNRAS 492 4716 2020* — *2020*

8. **The Splash without a merger** — **ApJ *Letters***
*J. A. S. Amarante, **LBeS**, V. P. Debattista, & M. C. Smith, ApJ Letters 891 L30 2020* — *2020*

7. **The Discreteness-driven Relaxation of Collisionless Gravitating Systems: Entropy Evolution and the Nyquist-Shannon Theorem** — **ApJ**
***LBeS**, W. de Siqueira Pedra, M. Valluri, ApJ 872 20 2019* — *2019*

6. **The Discreteness-driven Relaxation of Collisionless Gravitating Systems: Entropy Evolution in External Potentials, N-dependence, and the Role of Chaos** — **ApJ**
***LBeS**, W. de Siqueira Pedra, M. Valluri, L. Sodré, J.-B. Bru, ApJ 870 128 2019* — *2019*

5. **The Arrow of Time in the Collapse of Collisionless Self-gravitating Systems: Non-validity of the Vlasov-Poisson Equation during Violent Relaxation** — **ApJ**
***LBeS**, W. de Siqueira Pedra, L. Sodré, E. L. Perico, M. Lima, ApJ 846 125 2017* — *2017*

4. **Anisotropic q-Gaussian 3D velocity distributions in L-CDM haloes** — **MNRAS**
***LBeS**, G. A. Mamon, M. Duarte, R. Wojtak, S. Peirani, G. Boué, MNRAS 452 944 2015* — *2015*

3. **Statistical mechanics of self-gravitating systems: Mixing as a criterion for indistinguishability, PRD 90l3004 2014** — **PRD**
***LBeS**, M. Lima, L. Sodré, J. Perez,* — *2014*

2. **Testing phenomenological and theoretical models of dark matter density profiles with galaxy clusters** **MNRAS**
__LBeS__, M. Lima, L. Sodré, MNRAS 436, 2616 2013 *2013*

1. **Constraints on enhanced dark matter annihilation from IceCube results** **PRD**
I. F. M. Albuquerque, __LBeS__, C. Pérez de los Heros, PRD 85l3539 2012 *2012*

Teaching experience and training

- Attended the course "Postdoctoral Short Course on College Teaching in STEM" (10 weeks), at the Center for Research on Learning and Teaching, University of Michigan.
- Co-lectured the course "Hunting for the Dark: Black Holes and Dark Matter in the Milky Way" for high-school students in the Michigan Math and Science Scholars, 2023 (two weeks, w/ Monica Valluri).
- As a postdoc at University of Central Lancashire, I contributed to preparation and worked as Teaching Assistant (TA) for the course AA3853 Cosmology and Galaxies.
- As a graduate student, I worked as TA for several undergrad courses in the University of São Paulo, covering topics such as classical mechanics (basic and advanced levels), atomic and nuclear physics and electromagnetism.

Students supervision

- Alyssa Russell, University of Michigan, 2023, undergrad research project *Hunting hyper-velocity stars in the DESI MW survey*;
- Tigran Khachaturyants, University of Central Lancashire, 2018-2021 Ph.D. Thesis *The Milky Way's Warp in the Era of Gaia* (co-supervisor).

Other information

Selected Contributed Talks

Galactic bars: driving and decoding galaxy evolution, Granada, Spain: *2023*
54th Annual Division on Dynamical Astronomy of AAS Meeting, East Lansing, MI: *2023*
53rd Annual Division on Dynamical Astronomy of AAS Meeting New York, NY: *2022*
XLV Reunião Anual da Sociedade Astronômica Brasileira, Rio de Janeiro, Brasil: *2022*
National Astronomy Meeting, Bath, UK: **(invited)** *2021*
Galactic Disc and Halo Dynamics towards Gaia EDR3, Royal Astronomical Society, London, UK: *2020*
Collisionless Boltzmann Eq. and Modeling of Self-Gravitating Systems and Plasmas, Marseille, France: *2017*

Selected Colloquia and Seminars

University pf Colorado Boulder, Boulder, CO: *2023*
University of Florida (remote): *2023*
University of Alabama, Tuscaloosa, AL: *2023*
Institute of Astronomy, Cambridge, UK: *2019*
University of Central Lancashire, Preston, UK: *2019*
Basque Center of Applied Mathematics, Bilbao, Spain: *2017*
École Polytechnique, Laboratoire de Plasmas, Paris, France: *2017*
ICTP-SAIFR, São Paulo, Brasil: *2016*
University of São Paulo, Institute of Astronomy, São Paulo, Brasil: *2016*
University of São Paulo, Dept. of Mathematics, São Paulo, Brasil: *2016*
Institute d'Astrophysique de Paris, France: *2013*
DARK Cosmology Center, Copenhagen, Denmark: *2013*

Service

- Current co-chair of the MW Survey within the DESI collaboration;
- Former co-leader of the DM topical group within DESI MWS;
- Chair of NASA-ATP panel (Astrophysics Theory);
- SOC of the meeting "Great Lakes Clusters and Streams", Aug 21-24 (2023);
- SOC of the 54th meeting of the Division on Dynamical Astronomy os AAS (2023);
- Member of the Division on Dynamical Astronomy (DDA) of the AAS, Sociedade Brasileira de Física (SBF), and Sociedade Astronômica Brasileira (SAB);
- Refereed papers for ApJ, ApJ *Letters*, MNRAS, New Astronomy, Galaxies, and Revista Brasileira de Ensino de Fisica (Brazilian Journal for Physics Teaching);
- Member of the committee thesis of André Zamorano Vitorelli, entitled "Weak Gravitational Lenses: Measurement Techniques & Their Use as a Cosmic Dating System";
- Organized the Galaxies seminars at UCLan (2018-2021);
- Exposer in the outreach project "Moon at Harris museum" at Preston - UK, celebrating the 50th anniversary of the Moon landings;

Public codes

- `naif`: a pure-python package for numerical analysis of frequencies;
- Silverman test: a jupyter-notebook to check for statistical significance of k or more peaks.